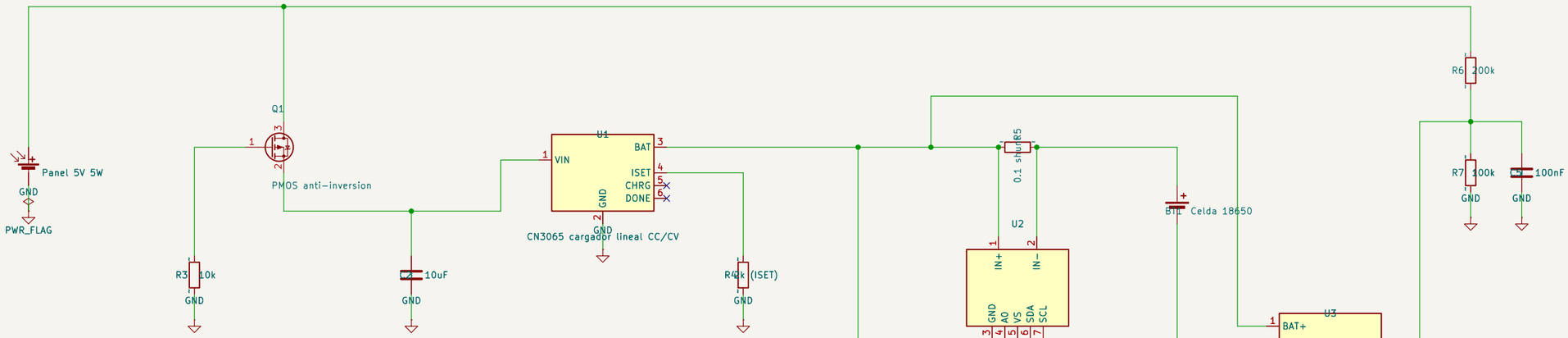
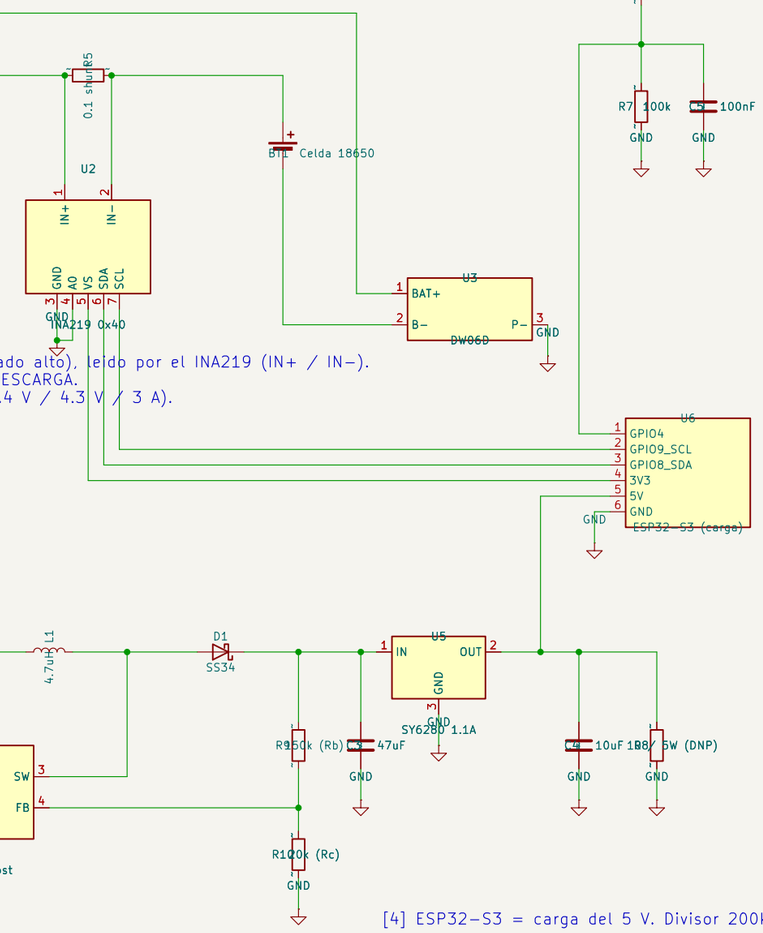
ESQUEMA 2/2 — ELECTRONICO CONCEPTUAL (equivalente interno de cada modulo)

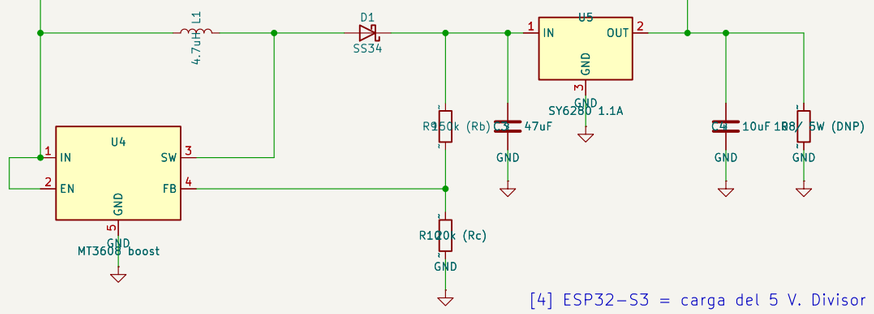
[1] PANEL + CARGADOR: PMOS anti-inversion + CN3065 lineal CC/CV (ISET 2k = 900 mA max).
Eficiencia lineal ~ Vbat/Vpanel (DFRobot: 73 % a 3.7 V / 900 mA). No es MPPT.



[2] MEDICION: shunt 0.1 ohm en BAT+ (lado alto), leido por el INA219 (IN+ / IN-).
Corriente positiva = CARGA, negativa = DESCARGA.
DW06D: FET serie en el negativo (corte 2.4 V / 4.3 V / 3 A).



[3] BOOST 5 V (MT3608): Vout = 0.6 x (1 + Rb/Rc) = 5.1 V. SY6280 limita a 1.1 A.
El nodo cuelga de VBAT_CHG: i_bat = i_cargador − i_carga.



[4] ESP32-S3 = carga del 5 V. Divisor 200k/100k (3:1) a GPIO4 + 100 nF.

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